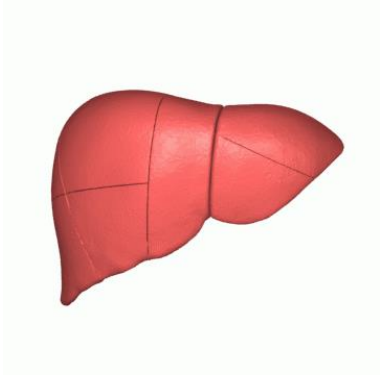




Physiology of The Liver

HMU/ LG4

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Learning outcome

By the end of this lecture, students should be able to:

- 1.Describe** the anatomical and functional organization of the liver.
- 2.Explain** the liver's role in carbohydrate, lipid, and protein metabolism.
- 3.Discuss** bile formation and bilirubin metabolism with clinical correlations.
- 4.Interpret** basic liver function tests in the context of liver physiology.

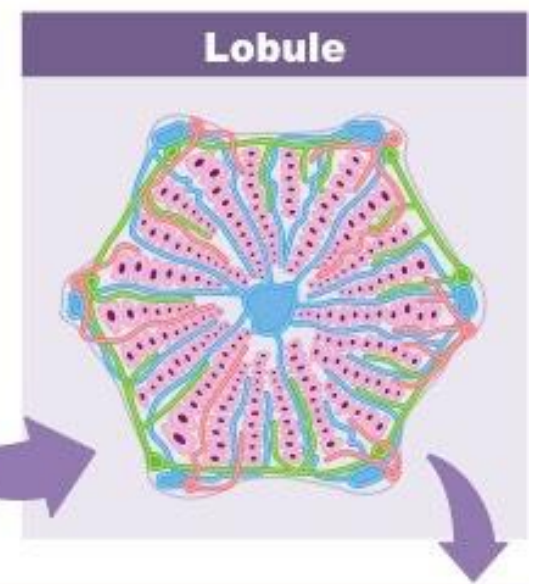
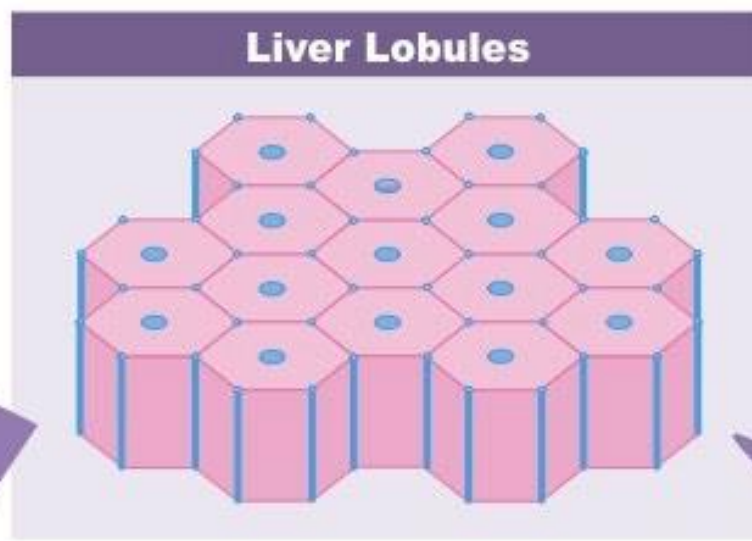
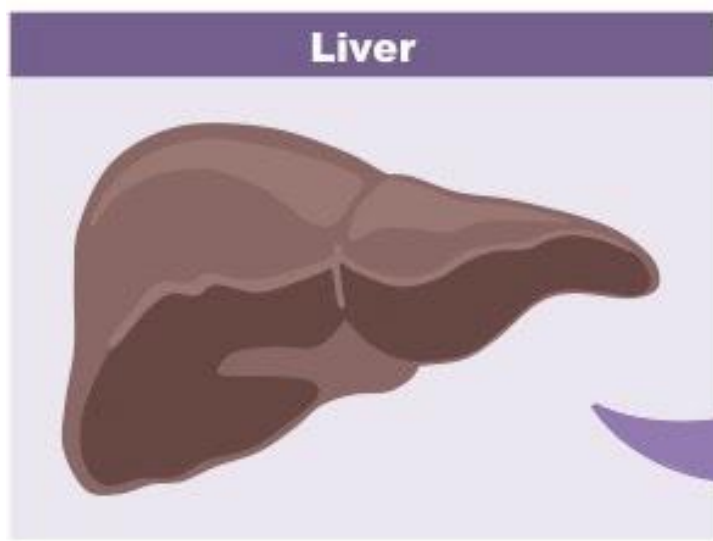
Physiology of the Liver

Introduction

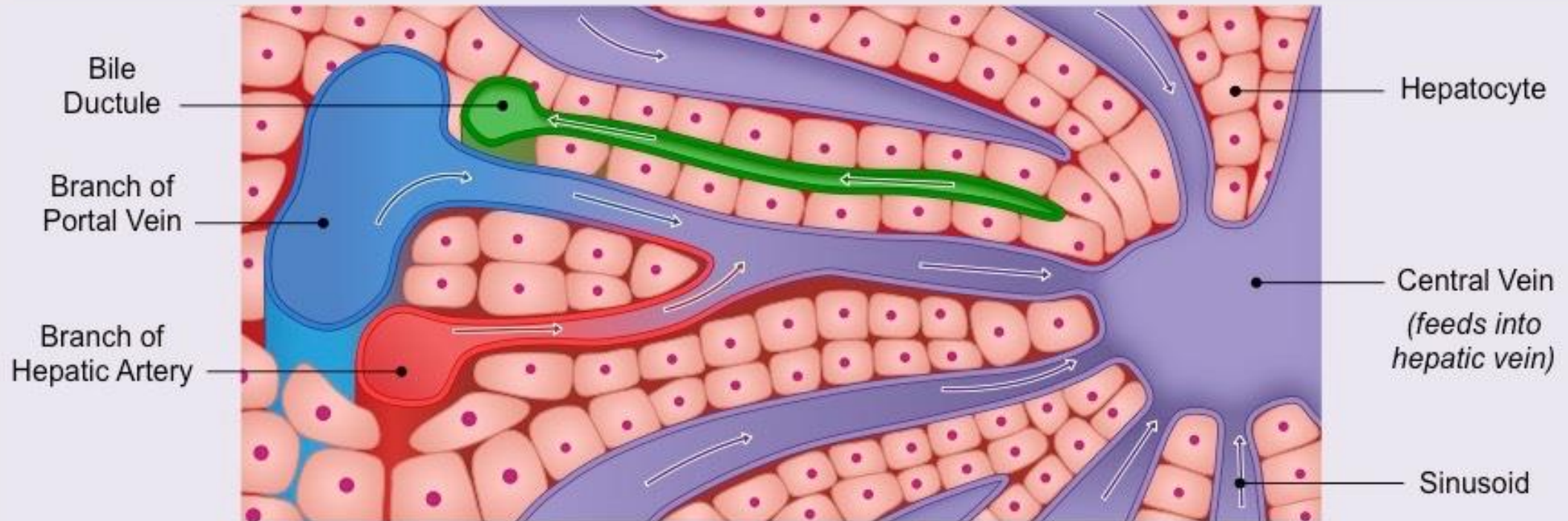
- The liver is the **largest internal organ** (≈ 1.5 kg in adults).
- Functions as a **metabolic hub**, processing nutrients, **detoxifying** substances, and **producing** vital proteins.
- Receives **dual blood supply**:
 - **Portal vein** (70–75%): nutrient-rich, oxygen-poor blood from GI tract.
 - **Hepatic artery** (25–30%): oxygen-rich blood from systemic circulation.
- Blood drains via hepatic veins into the inferior vena cava.

Structural & Functional Organization

- **Lobular structure:** hexagonal lobules with central vein.
- **Portal triad:** hepatic **artery**, portal **vein**, bile **duct**.
- **Sinusoids:** fenestrated capillaries allowing exchange between hepatocytes and blood.
- **Kupffer cells:** resident macrophages for immune defense.
- **Bile canaliculi:** channels for bile secretion.



Cross-Section of a Liver Lobule



Major **Functions** of the Liver

A. Metabolism

1. Carbohydrate Metabolism

1. Glycogenesis, glycogenolysis, gluconeogenesis.
2. Maintains blood glucose homeostasis (70–110 mg/dL).

Carbohydrate function the liver *does not perform*

- 1. It does not use glycogen/glucose primarily for its own energy needs during contraction (like muscle).**
- 2. It does not have the enzyme hexokinase in high amount (instead has glucokinase, with different regulation — so it cannot phosphorylate glucose at low concentrations the way brain/muscle can).**
- 3. It does not store glycogen for its own use; liver glycogen is reserved for maintaining blood glucose homeostasis, not for liver ATP.**

2. Lipid Metabolism

- Synthesis of cholesterol, triglycerides, lipoproteins.
- β -oxidation of fatty acids for energy.
- Conversion of excess carbohydrates/proteins into fat.

What lipid metabolism function that can not be done by the liver ?

1. The liver cannot utilize ketone bodies for energy (due to lack of thiophorase).
2. The liver does not store fat long-term (that's adipose tissue).
3. The liver does not hydrolyze circulating triglycerides (no lipoprotein lipase).

3. Protein Metabolism

- Synthesis of plasma proteins: **albumin**, clotting factors (II, VII, IX, X), transport proteins.
- Urea cycle: detoxifies ammonia from amino acid breakdown.
- Clinical correlation: **hypoalbuminemia → edema/ascites; coagulopathy** in liver failure.

Note that !!

- **The liver cannot produce immunoglobulins (done by plasma cells).**
- **It does not synthesize Factor VIII (produced in endothelial cells).**
- **It does not store protein long-term (muscle does).**

AGAIN ...

-  **What the liver *does* produce**
- Most plasma proteins: **albumin, fibrinogen, clotting factors (II, VII, IX, X, XI, V), binding proteins (transferrin, ceruloplasmin, haptoglobin).**
- Components of the **complement system (C1–C9)**, which are important for immunity.
-  **What the liver does *not* produce**
- **Immunoglobulins (antibodies, IgG, IgM, IgA, IgE, IgD) are not produced by hepatocytes.**
- They are synthesized by **plasma cells**, which are terminally differentiated **B-lymphocytes**, mainly in the **lymphoid organs** (spleen, lymph nodes, bone marrow, mucosal tissues).



⚠️ Why the confusion?

- In the **fetal liver**, hematopoiesis occurs (blood cell formation up to ~6 months gestation). That includes B-cell precursors, so sometimes textbooks mention “immune functions of the fetal liver.”
- But in the **adult**, the liver does **not make immunoglobulins**. It only produces **complement proteins** that *support* antibody function.

B. Storage

- Glycogen (glucose reserve).
- Fat-soluble vitamins: A, D, E, K.
- Vitamin B12 (several years' reserve).
- Iron (ferritin), copper.

- **C. Detoxification**

- Biotransformation of drugs and toxins (Phase I: oxidation, reduction; Phase II: conjugation).
- Breakdown of hormones (estrogen, cortisol, thyroid hormones).
- Clinical note: **drug toxicity in liver disease**, e.g., paracetamol overdose.

D. Bile Secretion

- Hepatocytes secrete bile (bile acids, cholesterol, phospholipids, bilirubin).
- Essential for **digestion & absorption of fats**.
- Bile salts aid micelle formation.
- Bilirubin metabolism:
 - From hemoglobin breakdown → conjugated in liver → excreted in bile.
 - Disorders: **jaundice** (pre-hepatic, hepatic, post-hepatic).

E. Immune & Endocrine Functions

- **Kupffer cells:** remove pathogens & debris.
- Modulates immune tolerance to gut antigens.
- Produces **IGF-1 (insulin-like growth factor)** under GH stimulation.
- Converts vitamin D to 25-hydroxycholecalciferol.

4. Liver Function Tests (LFTs)

- **Enzymes:** ALT, AST (hepatocellular damage); ALP, GGT (cholestasis).
- **Proteins:** Albumin (synthesis).
- **Bilirubin:** total and direct (conjugation/excretion).
- Clinical application: differentiation of hepatocellular vs cholestatic disease.

Clinical Correlations

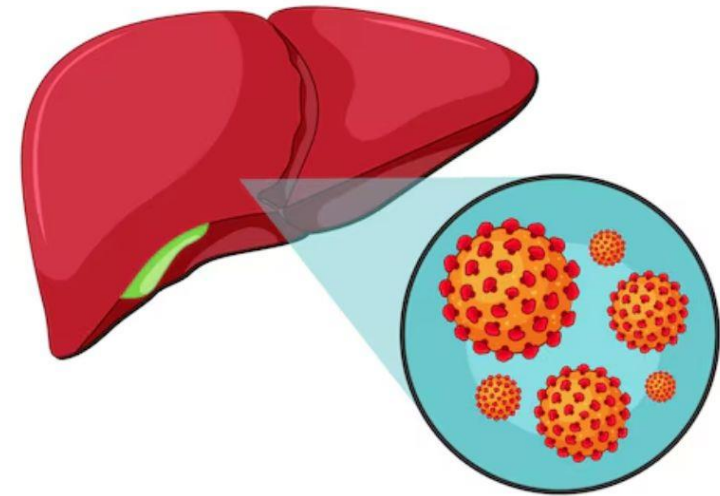
- **Cirrhosis:** chronic liver damage → fibrosis, portal hypertension, ascites.
- **Hepatic encephalopathy:** ammonia accumulation affects brain function.
- **Hepatitis:** viral, autoimmune, or toxic.
- **Hepatocellular carcinoma:** linked to HBV, HCV, cirrhosis.
- **Metabolic syndrome:** NAFLD as hepatic manifestation.

Causes of Liver Damage

1. Infections

- **Viral hepatitis:** Hepatitis A, B, C, D, E.
- Chronic HBV and HCV → cirrhosis, hepatocellular carcinoma.
- Other infections: malaria, leptospirosis, schistosomiasis (in endemic areas).

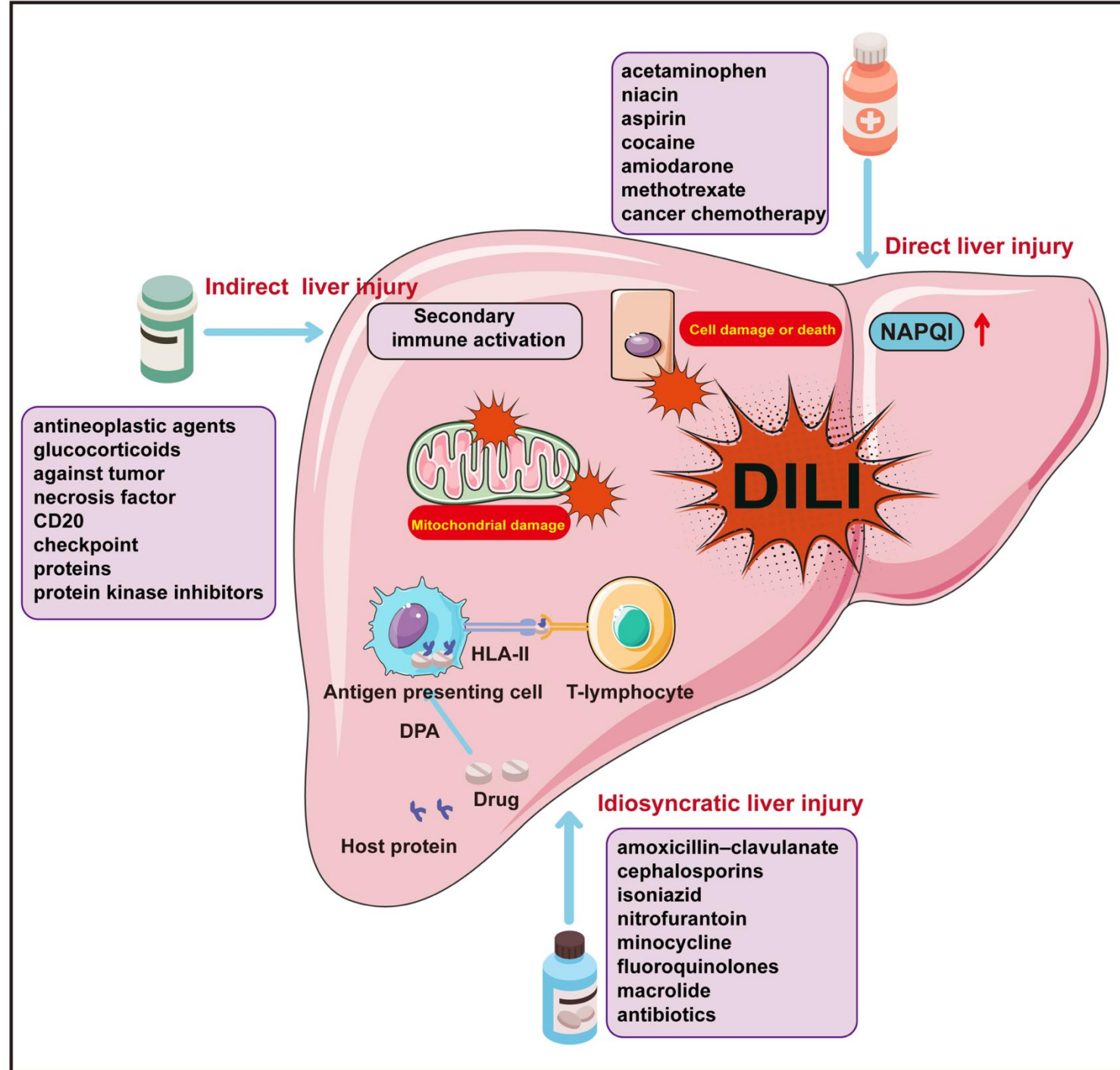
Hepatitis



Hepatitis is an inflammation of the liver, often caused by a virus. It can lead to fibro-sis (scarring) cirrhosis, or liver cancer.

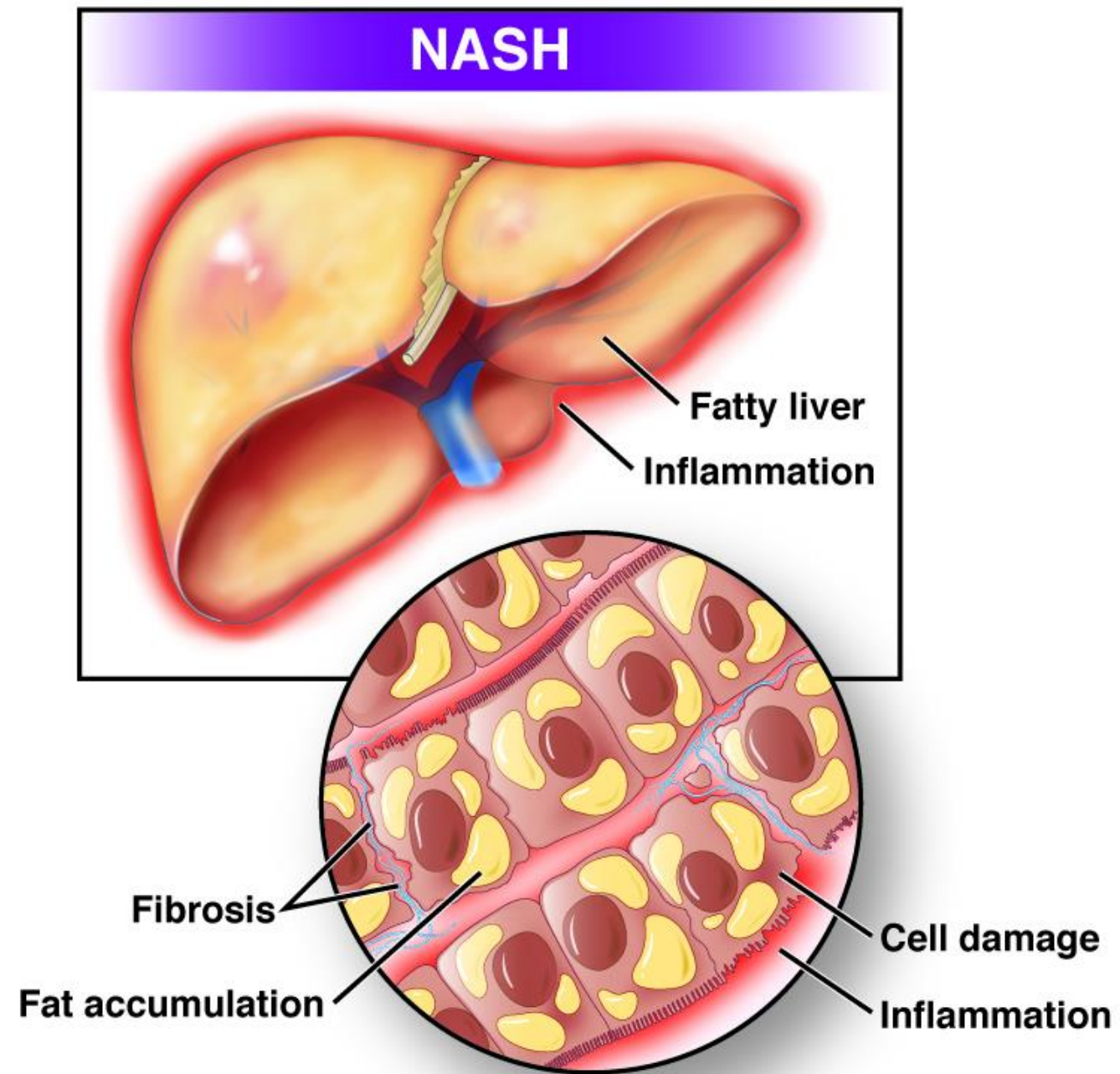
2. Toxins & Drugs

- **Alcohol:** fatty liver → alcoholic hepatitis → cirrhosis.
- **Drugs:** paracetamol overdose (most common cause of acute liver failure), anti-TB drugs (isoniazid, rifampin), methotrexate, amiodarone.
- **Herbal/toxic substances:** aflatoxin, certain mushrooms (*Amanita phalloides*).



3. Metabolic & Genetic Disorders

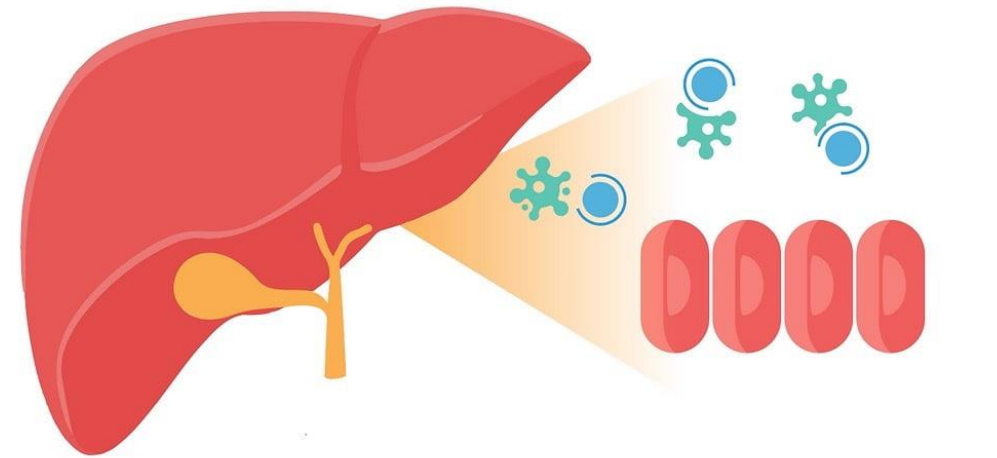
- **Non-alcoholic fatty liver disease (NAFLD):** linked to obesity, diabetes, metabolic syndrome.
- **Hemochromatosis:** excess iron deposition.
- **Wilson's disease:** copper accumulation.
- **Alpha-1 antitrypsin deficiency:** abnormal protein buildup in hepatocytes.
- **Glycogen storage diseases** (rare).



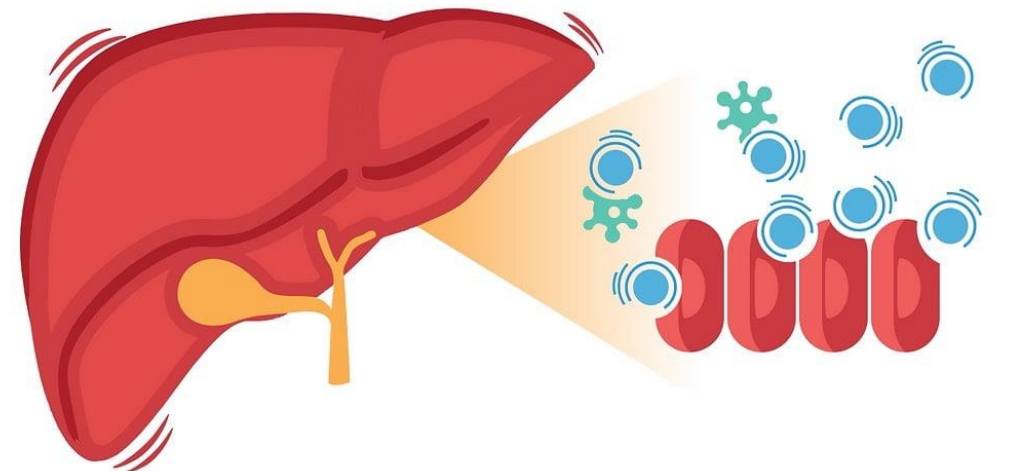
4. Immune-Mediated Damage

- **Autoimmune hepatitis:** body attacks hepatocytes.
- **Primary biliary cholangitis:** autoimmune destruction of bile ducts.
- **Primary sclerosing cholangitis:** fibrosis and strictures of bile ducts (linked with ulcerative colitis).

healthy liver

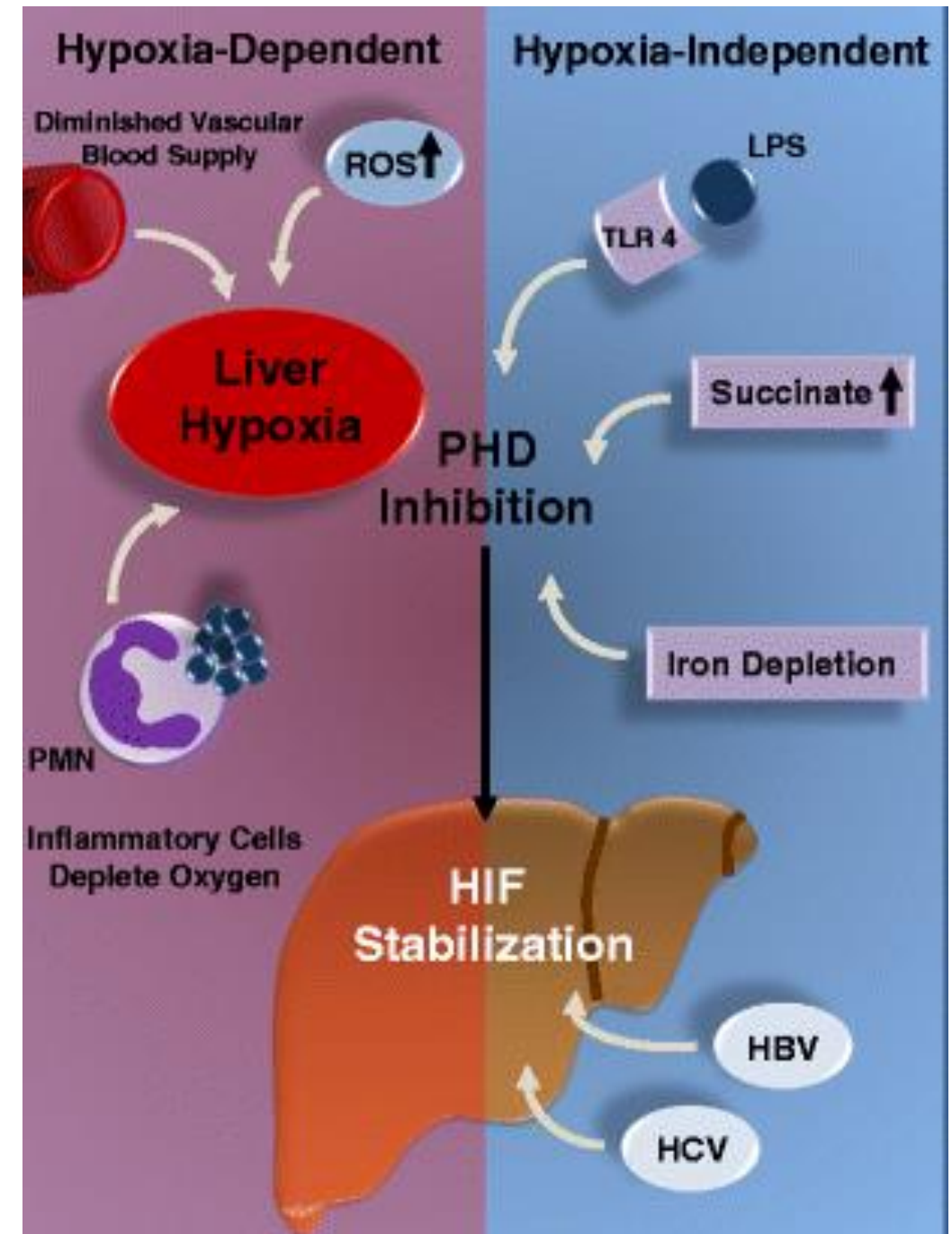


liver of an AIH patient



5. Vascular & Circulatory Causes

- **Ischemia** (shock liver): low blood flow/oxygen.
- **Budd–Chiari syndrome**: hepatic vein thrombosis.
- Right-sided heart failure → passive congestion of liver.



6. Neoplastic Conditions

- **Hepatocellular carcinoma (HCC):** often from chronic hepatitis/cirrhosis.
- **Metastases:** liver is a common site of secondary tumors (colon, pancreas, breast, lung).



